

⚖️ XAI: Explainability for Fairness & the Right to be Forgotten

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

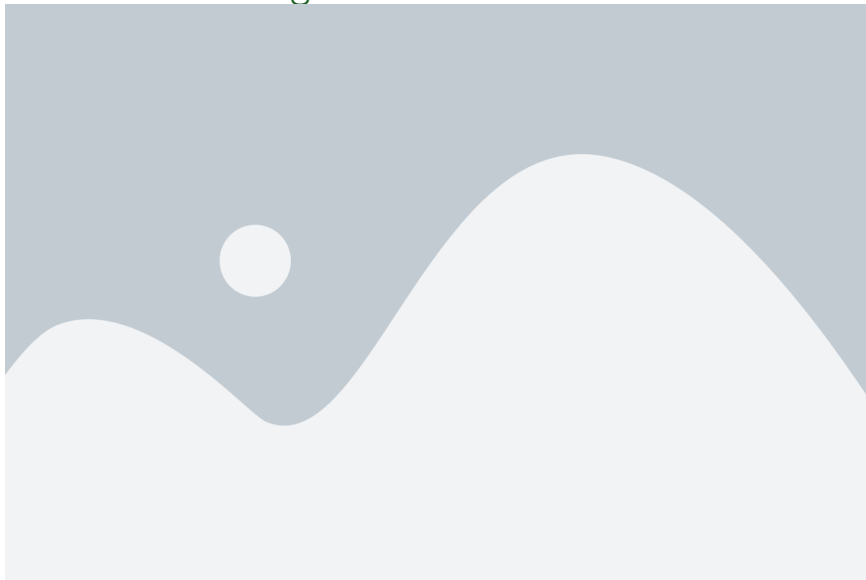
Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

🍷 Paper 1: Explainability for Fair Machine Learning ---



🍷 Fair Machine Learning ---



🍷 Challenges: Defining Fairness ---

Defining fairness is hard:

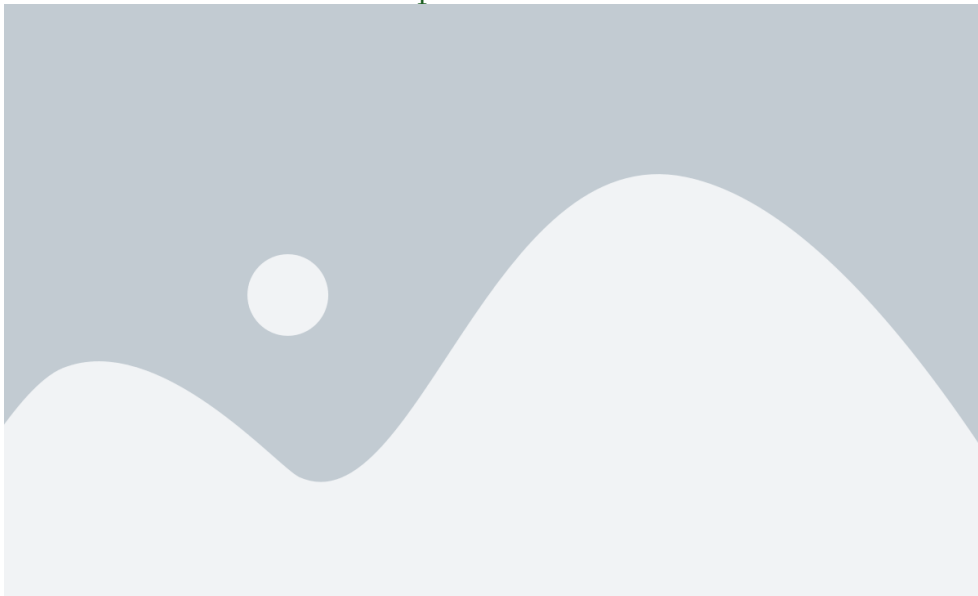
- Many competing definitions — statistics-based, causal-reasoning-based, etc.
- Group outcomes vs. individual outcomes
- Requires contextual understanding



🍷 Fairness Definition: Demographic Parity ---



🍷 Fairness Definition: Equalized Odds ---



🍲 Challenge: Explanation Methods Can Be Manipulated ---



Proposed Solution ---

Unified approach via Fairness Shapley Values

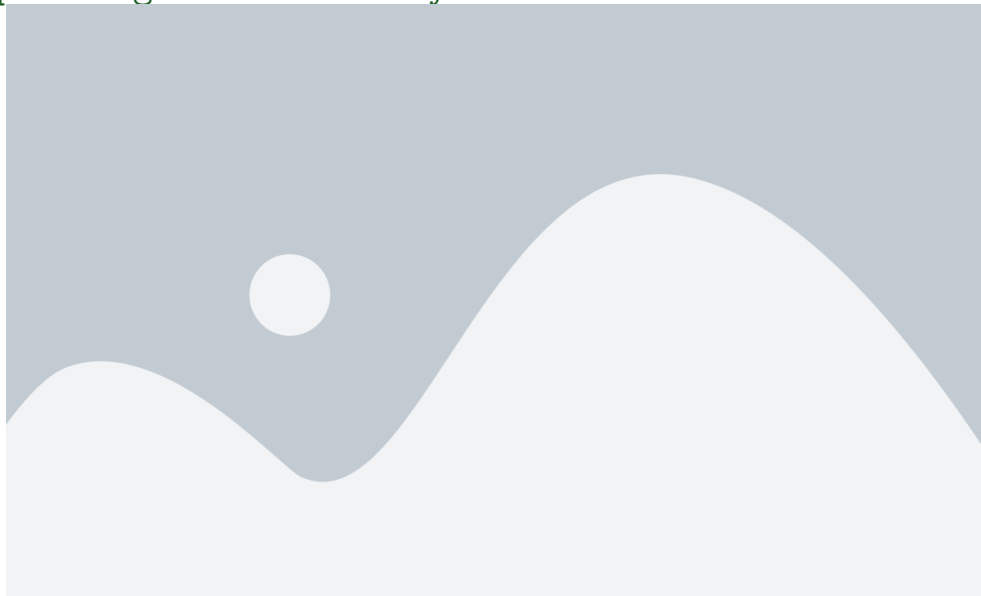
A single framework covering many group-fairness criteria:

- demographic parity, equalised odds, conditional demographic parity
- for each fairness definition, define a Shapley value function that attributes overall unfairness to individual features
- **Cannot hide unfairness** by manipulating explanations
 - ▶ Fairness Shapley values must collectively sum to the chosen fairness metric
- **Meta Algorithm**: learn a corrective perturbation δ_θ to the original model — no full retraining needed

(Begley et al. 2020)

Methodology ---

🍷 Explaining Model Accuracy ---



🍷 Explaining Model Fairness ---



🍷 Learning Corrective Perturbations ---



🍷 Experiments & Results ---

Datasets ---

****Adult Dataset**** (UCI)

Task: predict whether an individual earns more than \$50K/year based on demographics

****Sensitive attribute:**** sex

****COMPAS Recidivism Dataset**** (Larson et al., 2016)

Task: predict recidivism risk based on demographics

****Sensitive attribute:**** race

🍷 Explainability Results ---



🍷 Robustness of Fairness Explanation ---



🍷 Learnt Perturbations: Performance ---

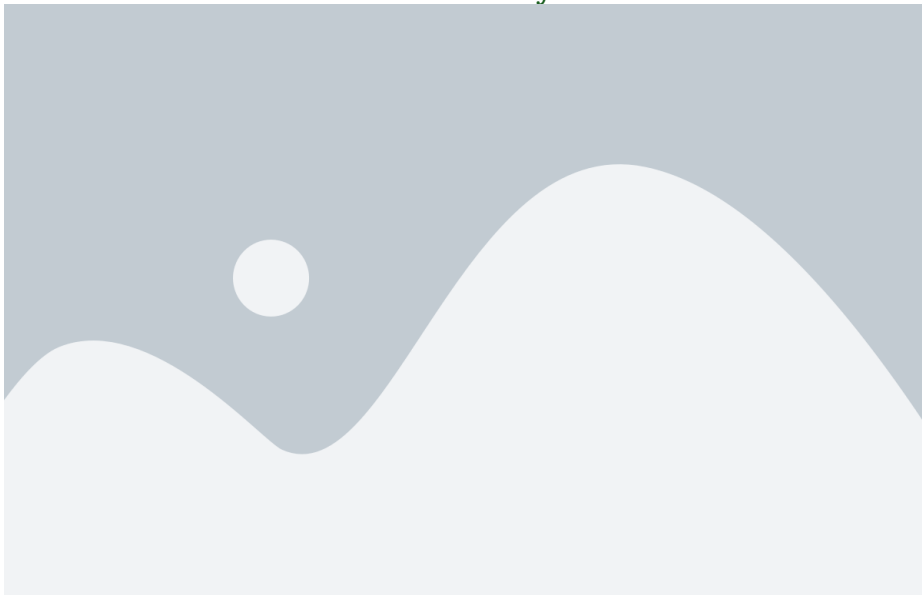


Demographic Parity: No significant accuracy reduction while imposing fairness constraint



Equalised Odds: Same result — perturbative approach competitive with full retraining

🍷 Learnt Perturbations: Stability ---

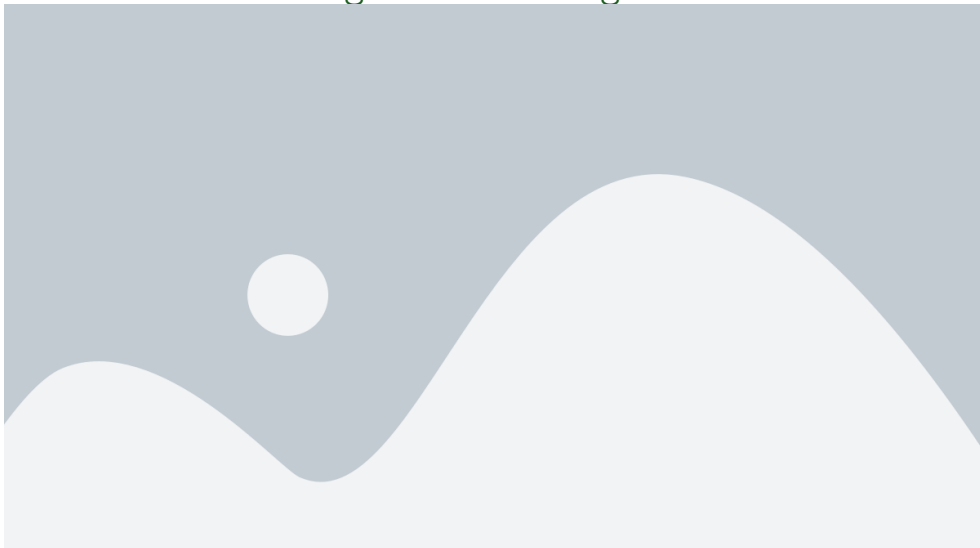


Discussion Questions: Fair ML ---

- ➊ What are the advantages and disadvantages of the perturbation method vs. other training-time fairness algorithms?
- ➋ If a feature contributes greatly to unfairness (e.g., marital status in demographic parity), is it always correct to remove it from the model?
- ➌ After reading this paper, what is your opinion on using interpretability methods to **validate** the fairness of ML models?

(Begley et al. 2020)

🍷 Paper 2: Towards Bridging the Gaps between the Right to Explanation and the Right to be Forgotten ---



🍷 Two Competing Rights ---

Right to an Explanation

Given a model prediction, provide a reason — via Algorithmic Recourse (counterfactuals)

Prior work: Multi-Objective CFE (Dandel et al. 2020), MACE (Karimi et al. 2020)

Right to be Forgotten

User can ask to have personal data removed from databases and models

Prior work: Descent-to-delete (Neel et al. 2021), Approximate deletion (Izzo et al. 2021)

The Conflict

Forgetting $\Rightarrow$ Model changes $\Rightarrow$ Counterfactual explanation becomes invalid $\Rightarrow$ Right to explanation not met

Previous Work & Motivation ---

- **Pawelczyk et al. 2022**: identified the tradeoff between right to explanation and right to be forgotten
 - ▶ Trade-off stems from the fact that current explanation methods ignore underlying model changes
- **ROAR** (Upadhyay et al. 2021): methods that assume certain model changes
 - ▶ But: we can't know *exactly* how a model changes as a result of forgetting
- Pawelczyk et al. highlighted the problem — there is a need for a **solution**

This Paper

ROCERF: first algorithmic framework to address this tradeoff, with theoretical guarantees

ROCERF: **RO**bust **C**ounterfactual **E**xplanations under the **R**ight to be **F**orgotten

- ① First framework to formally address the right-to-explanation vs. right-to-be-forgotten tradeoff
- ② Efficient approximation via first-order Taylor expansion (avoids $\binom{n}{k}$ retraining)
- ③ Theoretical bounds on cost and validity for linear and nonlinear models
- ④ ROCERF **outperforms** existing counterfactual explanation methods (SCFE, C-CHVAE, ROAR)

(Krishna, Ma, and Lakkaraju 2022)

ROCERF Framework ---

🍷 Notation ---



🍲 Counterfactual Explanation as Optimization ---



🍲 k -Removal Robust CFE ---

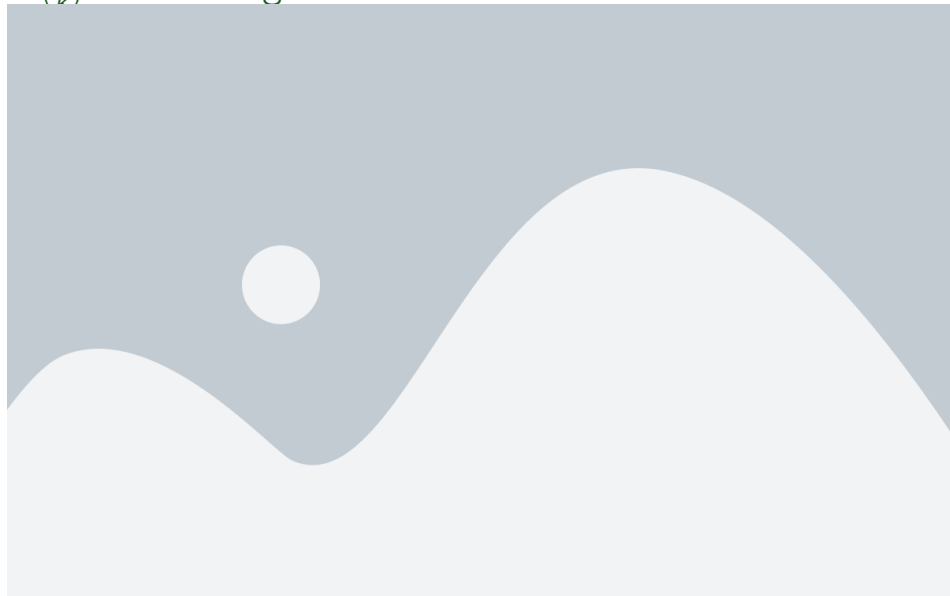


🍷 Efficient Approximation ---

Fix x and approximate $f_{\hat{\theta}_w}(x)$ via **first-order Taylor expansion** in w :

$$\tilde{f}_{\hat{\theta}_w}(x) = f_{\hat{\theta}_1}(x) + \left. \frac{\partial f_{\hat{\theta}_w}(x)}{\partial w} \right|_{w=1} (w - 1)$$

🍷 From $\binom{n}{k}$ to a Single Constraint ---



Practical Considerations ---

- **Computational cost:** $O(n)$ — only requires computing $\beta(x)^T H^{-1} g_i(\hat{\theta}_1)$ for each sample
- **Hyperparameters:** k (number of removals), δ (approximation error margin)
- **Linear models:** can avoid the backward pass entirely

$$\tilde{f}_{\hat{\theta}_w}(x) = \hat{\theta}_1^T x + \frac{1}{n} \sum_{i: w_i=0} x^T H^{-1} g_i(\hat{\theta}_1)$$

🍷 Theoretical Analysis: Validity & Cost ---



🍷 Experiments & Results ---

Experimental Setup ---

****Datasets:**** 3 real-world binary classification datasets from high-stakes decision making

- German Credit - COMPAS - Adult

****Models:**** regularized logistic regression + 3-layer feedforward NN

****Baselines:**** SCFE, C-CHVAE, ROAR

****Evaluation:**** - Validity: randomly remove fraction $\alpha \in [0.5\%, 5\%]$ - Repeat $M = 100$ times - $k = 0.5\%$ of training set, $\delta = 0$

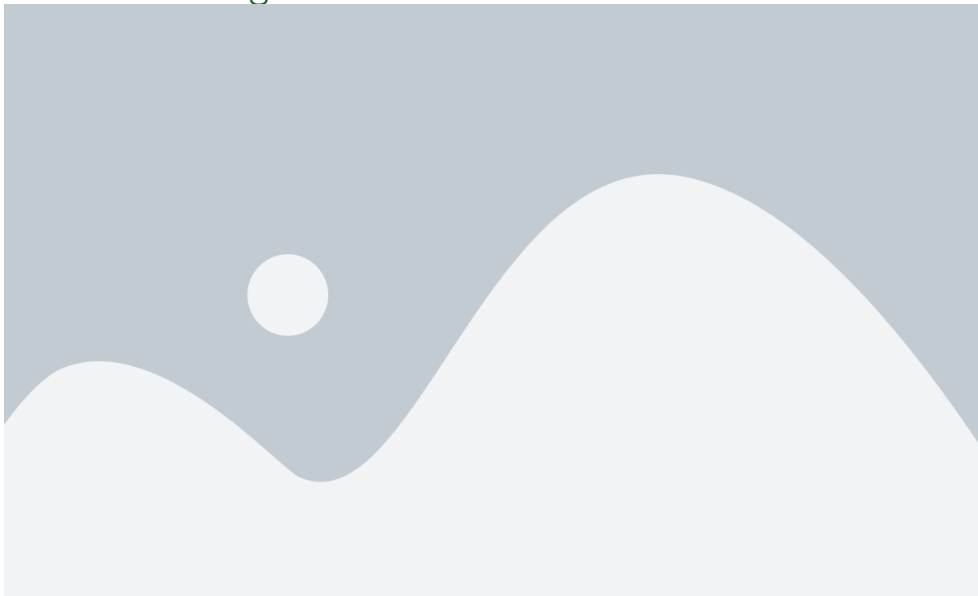
🍷 Results: Average Validity (Logistic Regression) ---



🍷 Results: Average Validity (Neural Networks) ---



🍲 Results: Average Cost ---



🍷 Sensitivity to Hyperparameter k ---



Conclusions: ROCERF ---

- **ROCERF** is the first framework to formally bridge the gap between the right to explanation and the right to be forgotten
- Provably robust to model updates triggered by data deletion requests
- Efficient: reduces exponential constraints to $O(n)$ computation via first-order approximation
- Theoretical guarantees on validity and cost for linear models; empirically strong for NNs

🍷 Limitations & Future Work ---

- **Non-linear models:** linear approximation may not capture complex NN behavior — validity can improve as more data is forgotten (non-intuitive)
- **ROAR vs. ROCERF:** results track closely in some settings — is ROAR sufficient?
- **Sensitivity to k :** why doesn't performance vary more with k ? — higher α regimes needed
- **Open question:** why has recourse literature focused on linear approximations when non-linear models exhibit non-intuitive behavior?

Discussion Questions: ROCERF ---

- ➊ This paper bridges the right-to-explanation vs. right-to-be-forgotten gap (unlike the privacy vs. explanation tradeoff, which seems more opposed). What other XAI tradeoffs do you think can be bridged?
- ➋ Given that ROAR exists, do we really need ROCERF? Could deletion of data parameters be viewed as a perturbation of model parameters?
- ➌ Why has recourse literature focused on linear approximations when results show non-linear models exhibit non-intuitive behavior?

(Krishna, Ma, and Lakkaraju 2022)

Thank You!

- Begley, Tom, Tobias Schwedes, Christopher Frye, and Ilya Feige. 2020. “Explainability for Fair Machine Learning.” *arXiv Preprint arXiv:2010.07389*.
- Krishna, Satyapriya, Jiaqi Ma, and Himabindu Lakkaraju. 2022. “Towards Bridging the Gaps Between the Right to Explanation and the Right to Be Forgotten.” In *Proceedings of the 39th International Conference on Machine Learning*.